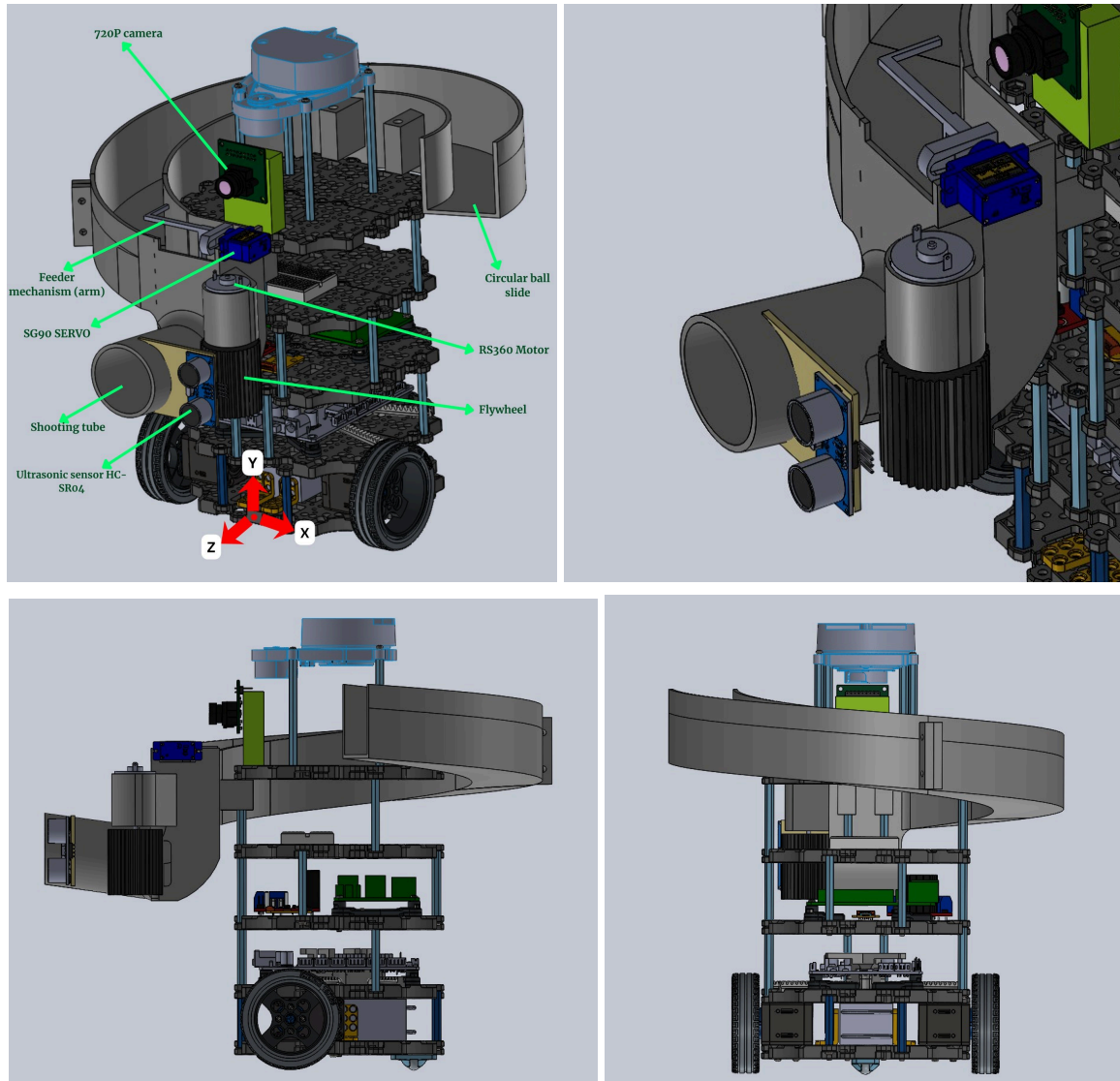


GROUP 2 USER MANUAL

Project Title: Autonomous Mobile Robots (AMRs) Intralogistics in a Smart IOT4.0 Warehouse

Course Code: CDE2310 Fundamentals of System Design



SPECIFICATIONS:

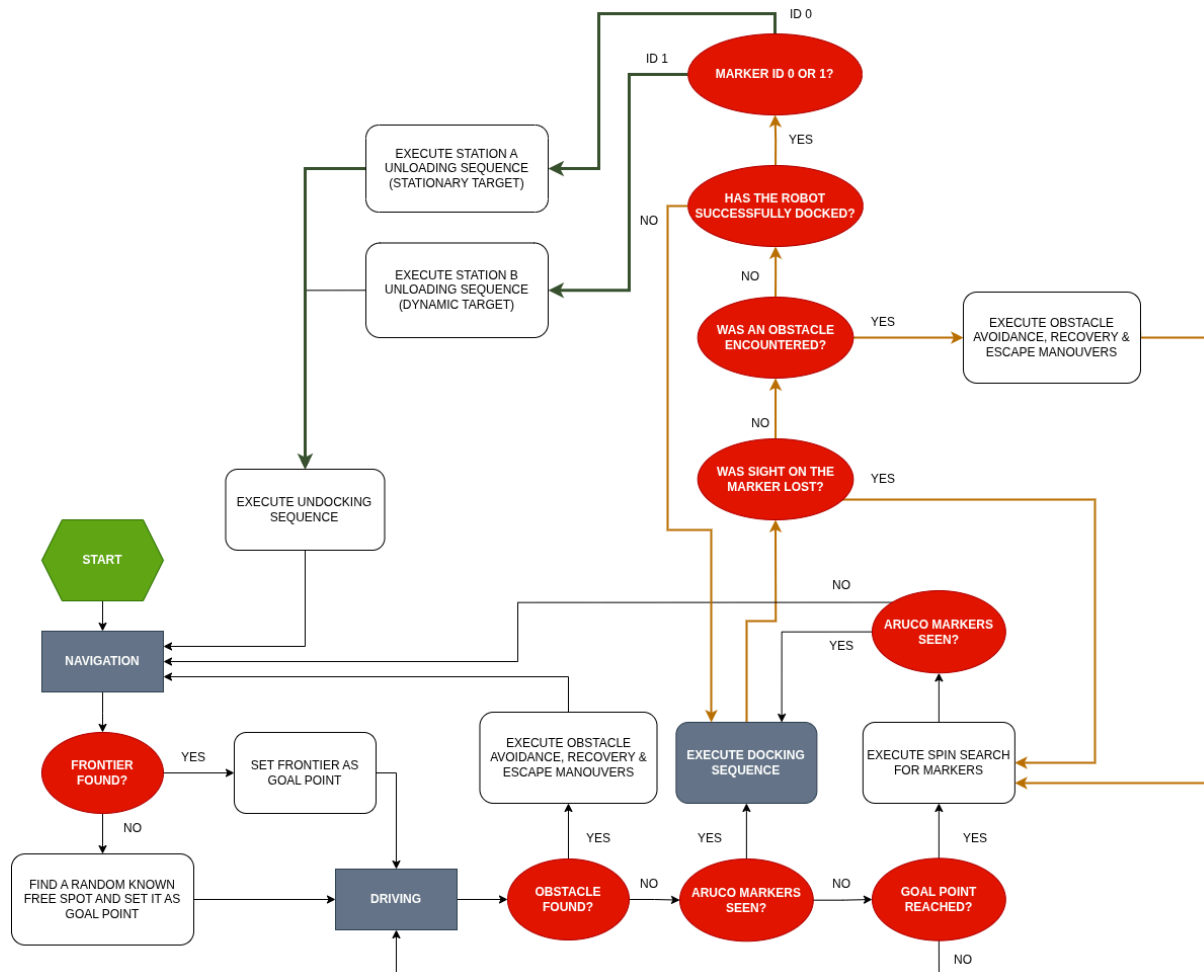
PLATFORM	Hardware: Turtlebot3 Software: Robot Operating System 2 Humble (ROS2)
SOFTWARE VERSION	Version 1.5.1
CENTER OF GRAVITY	-2.63mm (X), 122.43mm (Y), -53.58mm (Z)
WEIGHT	1785.5g (Fully Loaded)
BATTERY CAPACITY	1800mAh
BATTERY LIFE	Up to 1 Hour

CHARGING TIME	Half Charge: 1 Hour, Full Charge: 2.5 Hours
PAYLOAD CAPACITY	9 Ping Pong Balls

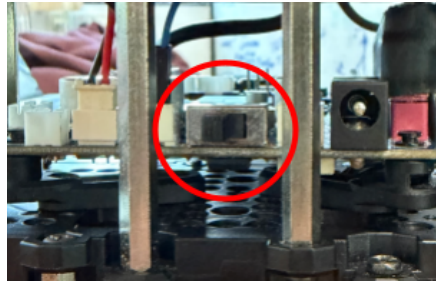
PURPOSE:

This AMR is to:

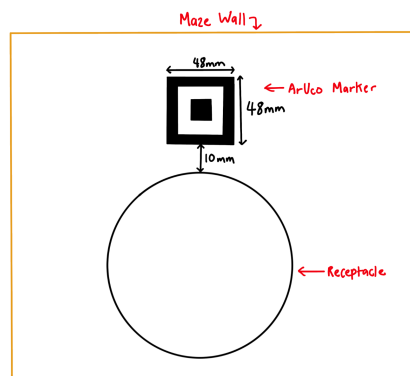
1. Self-navigate and traverse an unknown Warehouse Environment
2. Identify & Dock at Docking Stations via their respective ArUco Markers
3. Deliver the required payload batch into the receptacle at the station

SOFTWARE CONTROL FLOW**TECHNICAL GUIDE:****Operating Procedures:****ROBOT SET-UP**

1. Connect the Battery Wires to Supply Power to the Robot
2. Locate the On/Off Switch on the OpenCR (Second Layer of the Robot) to turn it on:



3. Place 9 Ping Pong Balls into the Circular Ball SlideMAZE SET-UP:
1. Prepare Two 5X5 250mm ArUco Markers with the following IDs:
 - a. ID 0: STATION A
 - b. ID 1: STATION B
2. Print Out the Markers with the following dimensions:
 - a. 48mm (W) x 48mm (H)
3. Place the Markers Above the Receptacle as seen below:



SOFTWARE SET-UP:

REMOTE LAPTOP	RASPBERRY PI 4B
Terminal Screen 1: Cartographer & RVis <code>rslam</code> Terminal Screen 2: Navigation & Docking <code>cd colcon_ws</code> <code>colcon build --packages-select auto_nav</code> <code>source install/setup.bash</code> <code>ros2 run auto_nav r2CDE2310_FINAL</code> When Prompted, In RVis, add the following publishers: 1. /goal_marker (RED DOT) 2. /lookahead_marker (BLUE DOT) 3. /planned_path (GREEN LINE) Terminal Screen 3: ArUco Marker Tracking <code>cd colcon_ws</code> <code>colcon build --packages-select ros2_aruco</code> <code>source install/setup.bash</code> <code>ros2 run ros2_aruco usbcam1</code>	Terminal Screen 1: Payload Delivery <code>cd cde2310_ws/src</code> <code>colcon build</code> <code>source install/setup.bash</code> <code>ros2 run rpilivecode rpilivecode</code> Terminal Screen 2: Camera Feed <code>cd turtlebot3_ws</code> <code>ros2 launch camera.launch.py</code> Terminal Screen 3: ROSBU <code>rosbu</code>

ACCEPTABLE DEFECTS LOG					
S/N	DESCRIPTION	POTENTIAL IMPACT	ASSESSMENT& Mitigations	IMPACT ON MISSION	STATUS
01	LiDAR Frontal Legs are not screwed in	LiDAR may be unstable during movement, leading to inaccurate SLAM	Hind Legs are tightly and securely screwed in, providing ample structural stability	None	Acceptable
02	OpenCR can be shifted around	Shifting may cause the on-off switch to be inaccessible and connections between the 12V Power Rail of the OpenCR and L298N may become loose	Degree of Shifting is minor and it is still able to stay in place during movement	None	Acceptable
03	When executing the command <code>"source install/setup.bash"</code> on the Raspberry Pi 4B, a warning which states <code>"ros2 not found: "/home/Group2/turtlebot3_ws/src/install/coin_d4_driver/share/coin_d4_driver/local_setup.bash"</code> appears	rpilivecode may fail to run, causing the robot to not be able to execute the delivery sequence	It was found that building within the src folder eliminates this issue	None	Acceptable
04	First time running rpilivecode after building, this message appears: <code>"/home/Group2/cde2310_ws/src/install/rpilivecode/lib/python3.10/site-packages/rpilivecode/rpilivecode.py:30: RuntimeWarning: This channel is already in use, continuing anyway. Use GPIO.setwarnings(False) to disable warnings. GPIO.setup(INT2, GPIO.OUT)"</code>	This warning informs users that the GPIO Pins being used have been initialised in a different state initially as to what is required. While this warning may not have an impact on the GPIO Pin's operations, it is a good practice to kill the program after running it the first time to allow for <code>GPIO.cleanup()</code> .	It is a good practice to kill the program after running it the first time to allow for <code>GPIO.cleanup()</code> .	Minimal	Acceptable

FACTORY ACCEPTANCE TEST:

TEST	STATUS	
	GO	NO GO
Raspberry Pi 4B Indicator Lights turn on when Powered		
LiDAR Spins Up		
OpenCR Start-Up Chime is heard		
Dynamixel Motors Move when the built-in diagnostic buttons on the OpenCR		

are pressed		
L298N Red Indicator Light is Illuminated		
Flywheel Motor Spins Freely without resistance when the wheel is physically rotated by hand		
All USB Cables are securely inserted into their respective ports		
USB Camera is recognised when the command ' <code>v4lt-ctl --list-devices</code> ' is run		
USB Camera Stream is started and a timer is set to trigger every 33ms		
The topic <code>/usbcam1_markers</code> publishes identified ArUco Marker(s) ID(s) and Position(s)		
SG90 Servo Snaps to Reset Position When ' <code>rpilivecode</code> ' is launched		

PARTS REPLACEMENT & MAINTENANCE LOG:

Date	Incident	Consequences	Rectifications	Reported By
15/03/26	Motor Driver did not power on upon startup	Motor was unable to spin	Replaced power wire	Udaiyan Bhan
14/04/26	Replacement of Prototype Flywheel Launcher with Final Design	Improved Reliability of Flywheel Launcher	NIL	Reagan Moses Widjaja
14/04/26	Flywheel Motor Wire Broke off during transfer to the final Launcher	Motor is unable to spin up when powered	Replaced with spare motor of the same type	Loo Zhong En Samuel
14/04/26	Rubber Wrapping of flywheel has become degraded with prior repeated testing	Flywheel may be unable to grip the Ping Pong Balls properly during delivery	Rubber Wrapping was replaced	Loo Zhong En Samuel